

13.0 DATA EEPROM MEMORY

The EEPROM data memory is readable and writable during normal operation (full V_{DD} range). This memory is not directly mapped in the register file space. Instead it is indirectly addressed through the Special Function Registers (SFRs). There are four SFRs used to read and write this memory. These registers are:

- EECON1
- EECON2 (Not a physically implemented register)
- EEDATA
- EEADR

EEDATA holds the 8-bit data for read/write and EEADR holds the address of the EEPROM location being accessed. PIC16F627A/628A devices have 128 bytes of data EEPROM with an address range from 0h to 7Fh. The PIC16F648A device has 256 bytes of data EEPROM with an address range from 0h to FFh.

The EEPROM data memory allows byte read and write. A byte write automatically erases the location and writes the new data (erase before write). The EEPROM data memory is rated for high erase/write cycles. The write time is controlled by an on-chip timer. The write time will vary with voltage and temperature, as well as from chip-to-chip. Please refer to AC specifications for exact limits.

When the device is code-protected, the CPU can continue to read and write the data EEPROM memory. A device programmer can no longer access this memory.

Additional information on the data EEPROM is available in the *PIC[®] Mid-Range Reference Manual* (DS33023).

REGISTER 13-1: EEDATA – EEPROM DATA REGISTER (ADDRESS: 9Ah)

R/W-x	R/W-x	R/W-x	R/W-x	R/W-x	R/W-x	R/W-x	R/W-x
EEDAT7	EEDAT6	EEDAT5	EEDAT4	EEDAT3	EEDAT2	EEDAT1	EEDAT0
bit 7							bit 0

bit 7-0 **EEDATn:** Byte value to Write to or Read from data EEPROM memory location.

Legend:

R = Readable bit W = Writable bit U = Unimplemented bit, read as '0'
 -n = Value at POR '1' = Bit is set '0' = Bit is cleared x = Bit is unknown

REGISTER 13-2: EEADR – EEPROM ADDRESS REGISTER (ADDRESS: 9Bh)

R/W-x	R/W-x	R/W-x	R/W-x	R/W-x	R/W-x	R/W-x	R/W-x
EADR7	EADR6	EADR5	EADR4	EADR3	EADR2	EADR1	EADR0
bit 7							bit 0

bit 7 **PIC16F627A/628A**
Unimplemented Address: Must be set to '0'

PIC16F648A
EEADR: Set to '1' specifies top 128 locations (128-255) of EEPROM Read/Write Operation
 bit 6-0 **EEADR:** Specifies one of 128 locations of EEPROM Read/Write Operation

Legend:

R = Readable bit W = Writable bit U = Unimplemented bit, read as '0'
 -n = Value at POR '1' = Bit is set '0' = Bit is cleared x = Bit is unknown

PIC16F627A/628A/648A

13.1 EEADR

The PIC16F648A EEADR register addresses 256 bytes of data EEPROM. All eight bits in the register (EEADR<7:0>) are required.

The PIC16F627A/628A EEADR register addresses only the first 128 bytes of data EEPROM so only seven of the eight bits in the register (EEADR<6:0>) are required. The upper bit is address decoded. This means that this bit should always be '0' to ensure that the address is in the 128 byte memory space.

13.2 EECON1 and EECON2 Registers

EECON1 is the control register with four low order bits physically implemented. The upper-four bits are non-existent and read as '0's.

Control bits RD and WR initiate read and write, respectively. These bits cannot be cleared, only set, in software. They are cleared in hardware at completion of the read or write operation. The inability to clear the WR bit in software prevents the accidental, premature termination of a write operation.

The WREN bit, when set, will allow a write operation. On power-up, the WREN bit is clear. The WRERR bit is set when a write operation is interrupted by a MCLR Reset or a WDT Time-out Reset during normal operation. In these situations, following Reset, the user can check the WRERR bit and rewrite the location. The data and address will be unchanged in the EEDATA and EEADR registers.

Interrupt flag bit EEIF in the PIR1 register is set when write is complete. This bit must be cleared in software.

EECON2 is not a physical register. Reading EECON2 will read all '0's. The EECON2 register is used exclusively in the data EEPROM write sequence.

REGISTER 13-3: EECON1 – EEPROM CONTROL REGISTER 1 (ADDRESS: 9Ch)

U-0	U-0	U-0	U-0	R/W-x	R/W-0	R/S-0	R/S-0
—	—	—	—	WRERR	WREN	WR	RD
bit 7							bit 0

bit 7-4 **Unimplemented:** Read as '0'

bit 3 **WRERR:** EEPROM Error Flag bit

- 1 = A write operation is prematurely terminated (any MCLR Reset, any WDT Reset during normal operation or BOR Reset)
0 = The write operation completed

bit 2 **WREN:** EEPROM Write Enable bit

- 1 = Allows write cycles
0 = Inhibits write to the data EEPROM

bit 1 **WR:** Write Control bit

- 1 = initiates a write cycle. (The bit is cleared by hardware once write is complete. The WR bit can only be set (not cleared) in software.)
0 = Write cycle to the data EEPROM is complete

bit 0 **RD:** Read Control bit

- 1 = Initiates an EEPROM read (read takes one cycle. RD is cleared in hardware. The RD bit can only be set (not cleared) in software).
0 = Does not initiate an EEPROM read

Legend:

R = Readable bit

W = Writable bit

U = Unimplemented bit, read as '0'

-n = Value at POR

'1' = Bit is set

'0' = Bit is cleared

x = Bit is unknown

13.3 Reading the EEPROM Data Memory

To read a data memory location, the user must write the address to the EEADR register and then set control bit RD (EECON1<0>). The data is available, in the very next cycle, in the EEDATA register; therefore it can be read in the next instruction. EEDATA will hold this value until another read or until it is written to by the user (during a write operation).

EXAMPLE 13-1: DATA EEPROM READ

```
BSF    STATUS, RP0    ;Bank 1
MOVLW  CONFIG_ADDR    ;
MOVWF  EEADR          ;Address to read
BSF    EECON1, RD      ;EE Read
MOVF   EEDATA, W       ;W = EEDATA
BCF    STATUS, RP0    ;Bank 0
```

13.4 Writing to the EEPROM Data Memory

To write an EEPROM data location, the user must first write the address to the EEADR register and the data to the EEDATA register. Then the user must follow a specific sequence to initiate the write for each byte.

EXAMPLE 13-2: DATA EEPROM WRITE

```
BSF    STATUS, RP0    ;Bank 1
BSF    EECON1, WREN    ;Enable write
BCF    INTCON, GIE     ;Disable INTs.
BTFS   INTCON, GIE     ;See AN576
GOTO   $-2
;
; Required Sequence
;
MOVLW  55h            ;
MOVWF  EECON2          ;Write 55h
MOVLW  AAh            ;
MOVWF  EECON2          ;Write AAh
BSF    EECON1, WR      ;Set WR bit
;begin write
BSF    INTCON, GIE     ;Enable INTs.
```

The write will not initiate if the above sequence is not followed exactly (write 55h to EECON2, write AAh to EECON2, then set WR bit) for each byte. We strongly recommend that interrupts be disabled during this code segment. A cycle count is executed during the required sequence. Any number that is not equal to the required cycles to execute the required sequence will cause the data not to be written into the EEPROM.

Additionally, the WREN bit in EECON1 must be set to enable write. This mechanism prevents accidental writes to data EEPROM due to errant (unexpected) code execution (i.e., lost programs). The user should keep the WREN bit clear at all times, except when updating EEPROM. The WREN bit is not cleared by hardware.

After a write sequence has been initiated, clearing the WREN bit will not affect this write cycle. The WR bit will be inhibited from being set unless the WREN bit is set.

At the completion of the write cycle, the WR bit is cleared in hardware and the EE Write Complete Interrupt Flag bit (EEIF) is set. The user can either enable this interrupt or poll this bit. The EEIF bit in the PIR1 registers must be cleared by software.

13.5 Write Verify

Depending on the application, good programming practice may dictate that the value written to the Data EEPROM should be verified (Example 13-3) to the desired value to be written. This should be used in applications where an EEPROM bit will be stressed near the specification limit.

EXAMPLE 13-3: WRITE VERIFY

```
BSF    STATUS, RP0    ;Bank 1
MOVF   EEDATA, W
BSF    EECON1, RD      ;Read the
                        ;value written
;
;Is the value written (in W reg) and
;read (in EEDATA) the same?
;
SUBWF  EEDATA, W       ;
BTFS   STATUS, Z       ;Is difference 0?
GOTO   WRITE_ERR      ;NO, Write error
:      ;YES, Good write
:      ;Continue program
```

13.6 Protection Against Spurious Write

There are conditions when the device may not want to write to the data EEPROM memory. To protect against spurious EEPROM writes, various mechanisms have been built-in. On power-up, WREN is cleared. Also when enabled, the Power-up Timer (72 ms duration) prevents EEPROM write.

The write initiate sequence and the WREN bit together help prevent an accidental write during brown-out, power glitch or software malfunction.

PIC16F627A/628A/648A

13.7 Using the Data EEPROM

The data EEPROM is a high endurance, byte addressable array that has been optimized for the storage of frequently changing information (e.g., program variables or other data that are updated often). When variables in one section change frequently, while variables in another section do not change, it is possible to exceed the total number of write cycles to the EEPROM (specification D124) without exceeding the total number of write cycles to a single byte (specifications D120 and D120A). If this is the case, then an array refresh must be performed. For this reason, variables that change infrequently (such as constants, IDs, calibration, etc.) should be stored in Flash program memory.

A simple data EEPROM refresh routine is shown in Example 13-4.

Note: If data EEPROM is only used to store constants and/or data that changes rarely, an array refresh is likely not required. See specification D124.

EXAMPLE 13-4: DATA EEPROM REFRESH ROUTINE

```
BANKSEL    0X80           ;select Bank1
CLRF       EEADR           ;start at address 0
BCF        INTCON, GIE     ;disable interrupts
BTFSC      INTCON, GIE     ;see AN576
GOTO      $ - 2
BSF        EECON1, WREN    ;enable EE writes

Loop
BSF        EECON1, RD      ;retrieve data into EEDATA
MOVLW      0x55            ;first step of ...
MOVWF      EECON2          ;... required sequence
MOVLW      0xAA            ;second step of ...
MOVWF      EECON2          ;... required sequence
BSF        EECON1, WR      ;start write sequence
BTFSC      EECON1, WR      ;wait for write complete
GOTO      $ - 1

#IFDEF __16F648A           ;256 bytes in 16F648A
    INCFSZ    EEADR, f      ;test for end of memory
#ELSE
    INCF      EEADR, f      ;next address
    BTFSS     EEADR, 7      ;test for end of memory
#ENDIF
;end of conditional assembly

GOTO      Loop            ;repeat for all locations

BCF        EECON1, WREN    ;disable EE writes
BSF        INTCON, GIE     ;enable interrupts (optional)
```

13.8 Data EEPROM Operation During Code-Protect

When the device is code-protected, the CPU is able to read and write data to the data EEPROM.

TABLE 13-1: REGISTERS/BITS ASSOCIATED WITH DATA EEPROM

Address	Name	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Value on Power-on Reset	Value on all other Resets
9Ah	EEDATA	EEPROM Data Register								xxxx xxxx	uuuu uuuu
9Bh	EEADR	EEPROM Address Register								xxxx xxxx	uuuu uuuu
9Ch	EECON1	—	—	—	—	WRERR	WREN	WR	RD	---- x000	---- q000
9Dh	EECON2 ⁽¹⁾	EEPROM Control Register 2								---- ----	---- ----

Legend: x = unknown, u = unchanged, - = unimplemented read as '0', q = value depends upon condition.
Shaded cells are not used by data EEPROM.

Note 1: EECON2 is not a physical register.